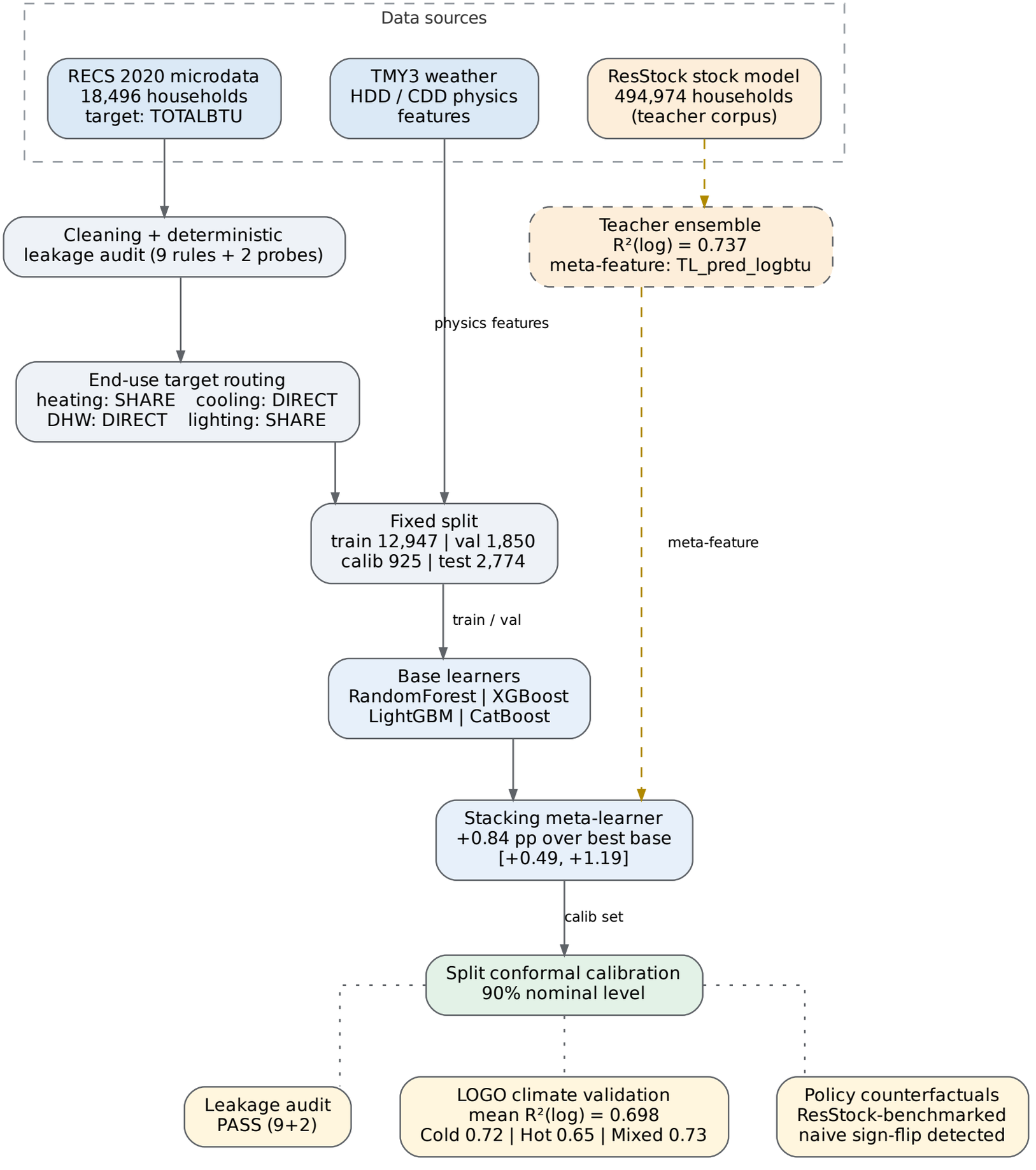
Data sources
RECS 2020 microdata
18,496 households
target: TOTALBTU
TMY3 weather
HDD / CDD physics
features
ResStock stock model
494,974 households
(teacher corpus)
Cleaning + deterministic
leakage audit (9 rules + 2 probes)
Teacher ensemble
R²(log) = 0.737
meta-feature: TL_pred_logbtu
physics features
End-use target routing
heating: SHARE cooling: DIRECT
DHW: DIRECT lighting: SHARE
Fixed split
train 12,947 | val 1,850
calib 925 | test 2,774
meta-feature
train / val
Base learners
RandomForest | XGBoost
LightGBM | CatBoost
Stacking meta-learner
+0.84 pp over best base
[+0.49, +1.19]
calib set
Split conformal calibration
90% nominal level
Leakage audit
PASS (9+2)
LOGO climate validation
mean R²(log) = 0.698
Cold 0.72 | Hot 0.65 | Mixed 0.73
Policy counterfactuals
ResStock-benchmarked
naive sign-flip detected